

# **BCG Data Science Project**

## **1. Project Name**

BCG Data Science Project – Customer Churn Analysis and Prediction

## **2. Project Description**

This project focuses on analysing customer churn and building a predictive machine-learning model to identify customers who are likely to leave an energy business. The work follows an end-to-end data science workflow, including exploratory data analysis (EDA), data preparation, feature engineering, model training and model evaluation.

## **3. Approach**

- Explored the customer dataset and identified important patterns and relationships.
- Performed data preparation and cleaning to create a usable modelling dataset.
- Combined customer-level information with aggregated pricing information.
- Created engineered pricing and consumption features to improve predictive information.
- Split the prepared dataset into training and testing data using a 75:25 split.
- Trained a Random Forest Classifier for customer churn prediction.
- Evaluated the model using accuracy, precision, recall and F1 score.

## **4. Technologies Used**

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Jupyter Notebook.

## **5. Model Evaluation**

| Metric    | Result |
|-----------|--------|
| Accuracy  | 90.39% |
| Precision | 82.61% |
| Recall    | 5.19%  |
| F1 Score  | 9.77%  |

Interpretation: Although accuracy and precision are relatively high, the low recall indicates that the model misses a large proportion of actual churners. Because churn is a minority class, accuracy should not be considered the only measure of model effectiveness.

## **6. Key Outcomes**

- The dataset contained 14,606 customers.
- The churn distribution was 13,187 retained customers (90.28%) and 1,419 churned customers (9.72%).
- The final modelling dataset contained 61 predictor features and one target variable.
- The data was divided into 10,954 training records and 3,652 testing records.
- Random Forest was used as the final classification model.
- The model achieved 90.39% accuracy and 82.61% precision.
- Recall was 5.19% and F1 score was 9.77%, highlighting the difficulty of identifying the minority churn class.

## 7. Challenges

- The churn dataset was imbalanced, with only 9.72% of customers classified as churned.
- Accuracy alone is insufficient for evaluating performance on an imbalanced classification problem.
- Customer and pricing data had to be combined into a suitable modelling dataset.
- Pricing and consumption features had to be engineered to improve the information available to the model.
- The model achieved high accuracy but very low recall, meaning that many actual churners were not identified.
- Model associations and feature importance must be interpreted carefully because association does not necessarily imply causation.

## 8. Business Recommendations

1. Develop targeted customer retention strategies using customer behaviour, pricing information and consumption patterns.
2. Monitor changes in off-peak, peak and mid-peak pricing variables and investigate customers experiencing significant price changes.
3. Use consumption\_change and consumption\_ratio to monitor meaningful changes in customer consumption behaviour.
4. Use churn predictions as a decision-support tool to prioritise customers, not as the sole basis for retention decisions because current recall is low.
5. Improve the model by addressing class imbalance, testing classification thresholds, evaluating class-weighted models and comparing additional algorithms.
6. Adopt proactive and personalised retention strategies based on customer-level insights.
7. Continuously monitor customer data, pricing data, churn rates and model performance.

## 9. Project Outcome

The project demonstrates an end-to-end customer churn analytics workflow, from data preparation and exploratory analysis through feature engineering and machine-learning evaluation. The results provide a foundation for data-driven customer retention, while also showing that further model improvement is required before churn predictions can be used as a reliable standalone system.

## GitHub Repository

[https://github.com/Mahanand4/BCG\\_Data\\_Science\\_Project](https://github.com/Mahanand4/BCG_Data_Science_Project)

## Certificate

[BCG Data Science Virtual Experience Certificate](#)